Bibliometric Credibility Weighting

h-index, Field Normalisation, and Citation Impact Indicators

Literature Review Reference Note · Compiled March 2026

1. Introduction

Bibliometric indicators are quantitative proxies for research quality and scholarly influence. Their use in evaluation contexts — grant allocation, tenure decisions, institutional rankings — has grown substantially since the 2000s, but so has awareness of their limitations. This note summarises the principal metrics, their theoretical bases, and the methodological considerations most relevant to credibility weighting in systematic literature review.

2. The h-index

Proposed by Hirsch (2005), the h-index is defined as the largest number *h* such that a researcher has at least *h* publications each cited at least *h* times. It combines productivity and impact into a single integer and is monotonically non-decreasing over a career.

2.1 Formal definition

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h = max{n : f(n) ≥ n} where f(n) = citations to the nth most-cited paper
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2.2 Known limitations

- Penalises early-career researchers because *h* is bounded by career length.
- Insensitive to citation counts above the threshold (a paper with 10,000 cites contributes no more than one with exactly *h*).
- Not comparable across disciplines without normalisation (see Section 3).
- Vulnerable to self-citation inflation (Bartneck & Kokkelmans, 2011).
- Does not account for author order or contribution fraction.

2.3 Variants

Variant	Description	Key reference
g-index	Largest g such that top-g papers have ≥ g² total cites	Egghe (2006)
m-quotient	h divided by academic age (years since first pub.)	Hirsch (2005)
i10-index	Number of papers with ≥ 10 citations	Google Scholar
AR-index	Square root of sum of annual citation rates in core	Jin et al. (2007)
hc-index	Age-weighted h-index discounting old papers	Sidiropoulos et al. (2007)

Table 1. Selected h-index variants and key references.

3. Field Normalisation

Citation norms vary substantially across disciplines. Biological sciences routinely produce papers with hundreds of citations; mathematics or philosophy papers may accumulate fewer than twenty yet be highly influential. Raw citation counts are therefore misleading for cross-field comparisons. Field normalisation addresses this by expressing a paper's impact relative to the expected citation rate for its field, publication year, and document type.

3.1 Field-Weighted Citation Impact (FWCI)

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FWCI = (actual citations) ÷ (expected citations for same field, year, document type)
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A FWCI of 1.0 equals the world average; 2.0 indicates twice the expected impact. Scores are computed by Elsevier's Scopus/SciVal platform using ~330 subject categories. The indicator is also referred to as the Crown Indicator (Waltman et al., 2011) in some contexts, though the two differ subtly in aggregation method.

3.2 Percentile-based normalisation

Because FWCI distributions are highly right-skewed, percentile ranks are often preferred. The Highly Cited Papers threshold (top 1% or 10% of field-year cohort) is used by Clarivate InCites and the National Science Foundation's Science & Engineering Indicators. Percentile ranks avoid the instability of ratio-based indicators when expected citation counts are very small, as frequently occurs in humanities and social science subfields (Waltman & van Eck, 2013).

3.3 Field definition challenges

The credibility of any normalised indicator depends critically on how 'field' is defined. Broad categories (e.g., 'Life Sciences') pool papers with very different citation dynamics; narrow categories may be too small for stable expected values. The choice of journal-based vs. paper-level classification further affects results. Waltman & van Eck (2012) provide a systematic comparison of classification systems and their impact on normalised indicators.

4. Journal-Level Indicators and Their Misuse

The Journal Impact Factor (JIF), introduced by Garfield (1955) and operationalised through the ISI Journal Citation Reports, measures the mean citation rate of a journal's articles in the two preceding years. It remains widely used but is problematic when applied to individual papers — a fallacy documented extensively in the literature.

- JIF distributions are highly skewed: a minority of papers account for the majority of citations (Seglen, 1997).
- JIF rewards review articles disproportionately over original research.
- JIF cannot be compared across disciplines without normalisation (same issue as raw h-index).
- The San Francisco Declaration on Research Assessment (DORA, 2012) and the Leiden Manifesto (Hicks et al., 2015) both explicitly recommend against using JIF as a proxy for individual paper quality.

The SCImago Journal Rank (SJR) and Source Normalised Impact per Paper (SNIP; Moed, 2010) are preferable alternatives. SNIP explicitly corrects for field-specific citation potential, making it more suitable for cross-disciplinary comparisons.

5. Advanced Citation Analysis

5.1 Smart citations and citation context

scite.ai classifies citations as supporting, contradicting, or mentioning, providing a richer credibility signal than citation counts alone. Nicholson et al. (2021) demonstrate that contradicting citations are systematically underweighted in standard bibliometric evaluations, creating a positive bias in perceived credibility.

5.2 Co-citation and bibliographic coupling

Co-citation analysis (Small, 1973) identifies intellectual structure by measuring how often pairs of papers are cited together. Bibliographic coupling (Kessler, 1963) links documents sharing common references. Both methods are used in systematic reviews to map knowledge domains and identify foundational versus peripheral literature without relying solely on citation counts.

5.3 Altmetrics

Altmetrics (Priem et al., 2010) aggregate non-traditional impact signals including social media mentions, news coverage, policy document references, and Wikipedia citations. While they capture societal reach not reflected in academic citations, they are susceptible to noise, algorithmic amplification, and disciplinary bias. Altmetric Attention Scores (AAS) should be used as supplementary rather than primary credibility indicators.

6. Normative Frameworks for Responsible Use

Framework	Year	Core recommendation
Leiden Manifesto	2015	Use multiple indicators; quantitative data should support, not supplant, qualitative judgement
DORA	2012	Do not use journal-level metrics (e.g., JIF) as a proxy for individual paper or researcher quali
Metric Tide Report	2015	Context-sensitive indicator choice; transparency in methodology
INORMS Declaration	2019	Responsible metrics literacy among evaluators is essential
Helsinki Initiative	2019	Multilingual and non-English publications must not be systematically disadvantaged

Table 2. Key normative frameworks for bibliometric use in evaluation.

7. Practical Guidance for Literature Review

When using bibliometric signals to weight sources in a systematic review, the following practices are recommended:

1. **Do not rank papers solely by citation count.** Use field-normalised indicators (FWCI, percentile rank) where cross-disciplinary comparison is required.

2. **Report the citation window.** Papers cited over a 3-year window are standard for FWCI; shorter windows favour recent work, longer windows favour established fields.
3. **Distinguish citation types where possible.** scite.ai or manual coding of supporting vs. contradicting citations adds credibility signal beyond counts.
4. **Use h-index as a researcher-level, not paper-level, signal.** When assessing source authority, h-index contextualises a researcher's career trajectory but should be compared within-field.
5. **State your data source.** FWCI values differ across Scopus, Web of Science, and Dimensions due to different coverage and field classifications.
6. **Apply a multi-indicator approach.** No single metric is sufficient. A combination of normalised citation impact, source quality indicator, and citation context provides the most defensible credibility weighting.

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